

Splunk Boss of the SOC V1 (Ransomware)

Scenario 2

Ransomware

After the excitement of yesterday, Alice has started to settle into her new job. Sadly, she realizes her new colleagues may not be the crack cybersecurity team that she was led to believe before she joined. Looking through her incident ticketing queue she notices a "critical" ticket that was never addressed. Shaking her head, she begins to investigate. Apparently on August 24th Bob Smith (using a Windows 10 workstation named we8105desk) came back to his desk after working-out and found his speakers blaring (click below to listen), his desktop image changed (see below) and his files inaccessible.

Alice has seen this before... ransomware. After a quick conversation with Bob, Alice determines that Bob found a USB drive in the parking lot earlier in the day, plugged it into his desktop, and opened up a word document on the USB drive called "Miranda_Tate_unveiled.dotm". With a resigned sigh she begins to dig into the problem...

Incident Summary

User connected an unknown USB drive and opened a malicious Word document which resulted in ransomware execution, file encryption, wallpaper change, and audio playback.

Objective:

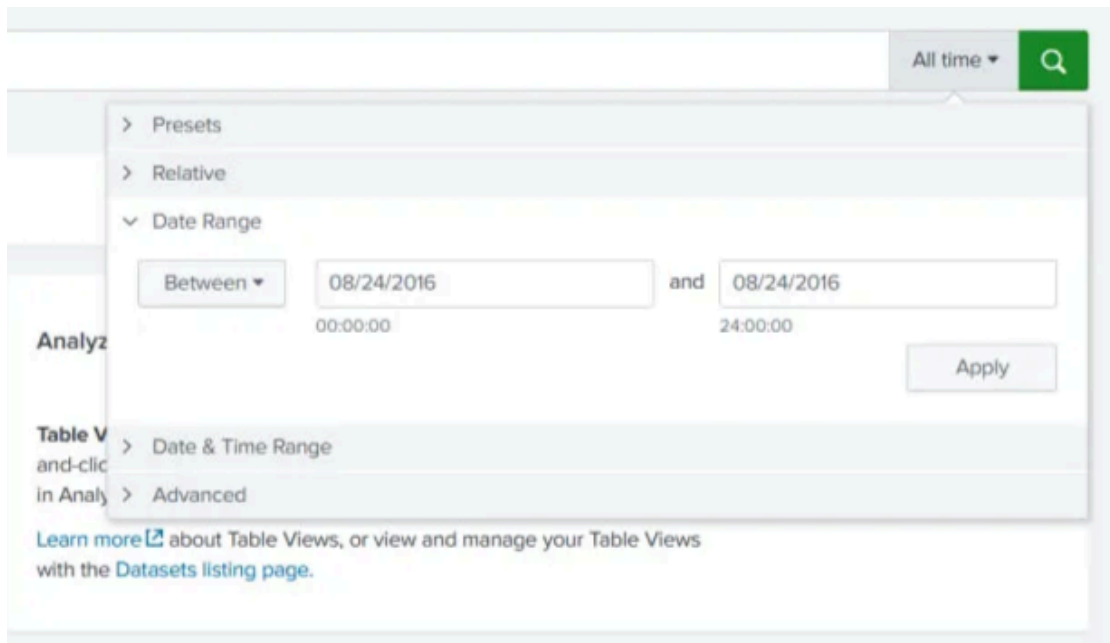
Investigate the incident, identify root cause, extract IOCs, and build a full attack timeline using Splunk.

Scope Definition

- Host: `we8105desk`
- User: `Bob Smith`
- Time Range: `August 24`
- Suspected File: `Miranda_Tate_unveiled.dotm`

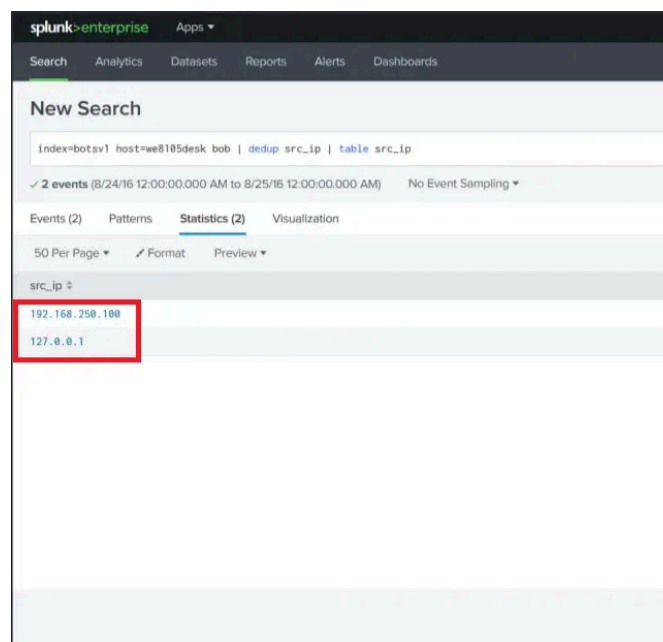
Setting the Time Range

I will first set the time Range to the time of incident.



Q1. What was the most likely IPv4 address of we8105desk on 24AUG2016 ?

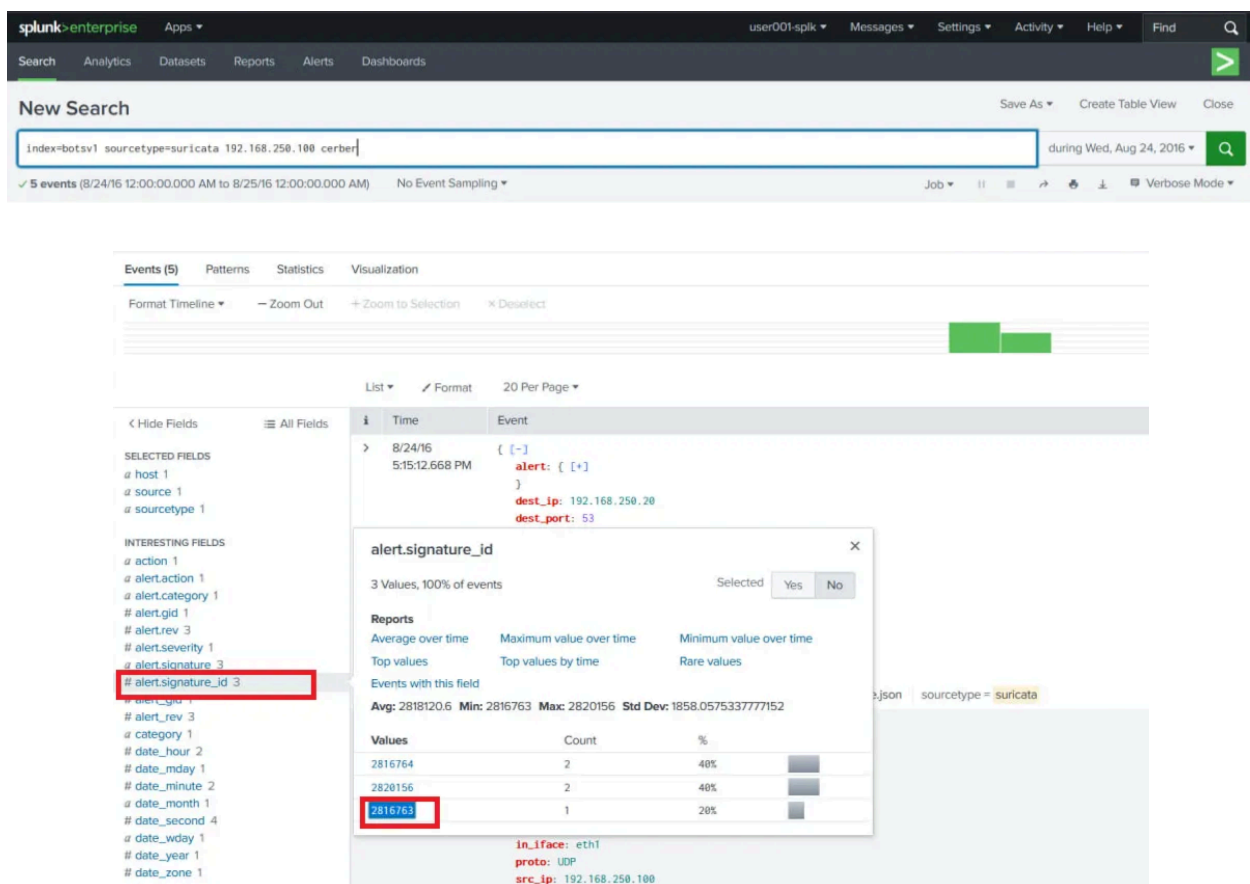
```
index=botsv1 host=we8105desk bob | dedup src_ip | table src_ip
```



Ans:192.168.250.100

Q2. Amongst the Suricata signatures that detected the Cerber malware, which one alerted the fewest number of times? Submit ONLY the signature ID value as the answer.

index=botsv1 sourcetype=suricata 192.168.250.100 cerberQ3. What fully qualified domain name (FQDN) does the Cerber ransomware attempt to direct the user to at the end of its encryption phase?



The screenshot shows the Splunk Enterprise interface. The search bar contains the query: `index=botsv1 sourcetype=suricata 192.168.250.100 cerber`. The search results show 5 events. A field picker for `alert.signature_id` is open, showing a table of values with the following data:

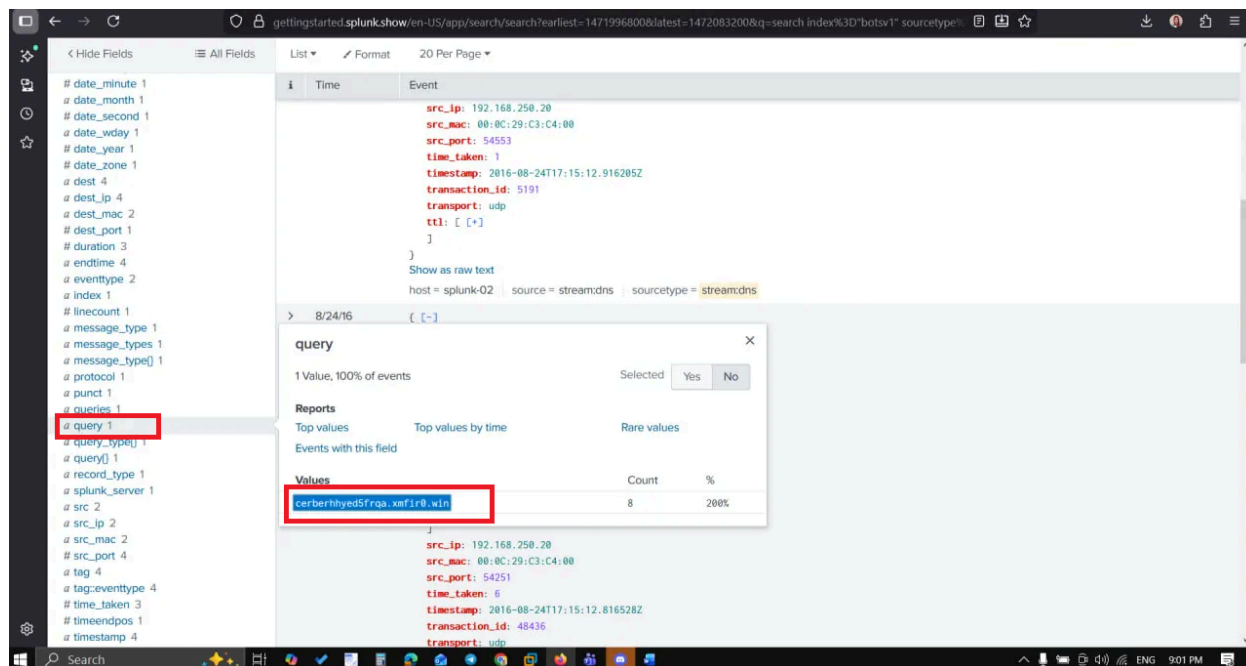
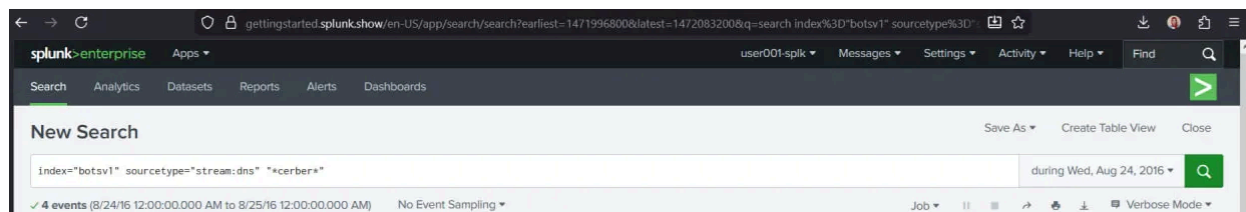
Values	Count	%
2816764	2	40%
2820156	2	40%
2816763	1	20%

The value `2816763` is highlighted in the table, indicating it is the signature ID that alerted the fewest number of times.

Ans:2816763

Q3. What fully qualified domain name (FQDN) does the Cerber ransomware attempt to direct the user to at the end of its encryption phase?

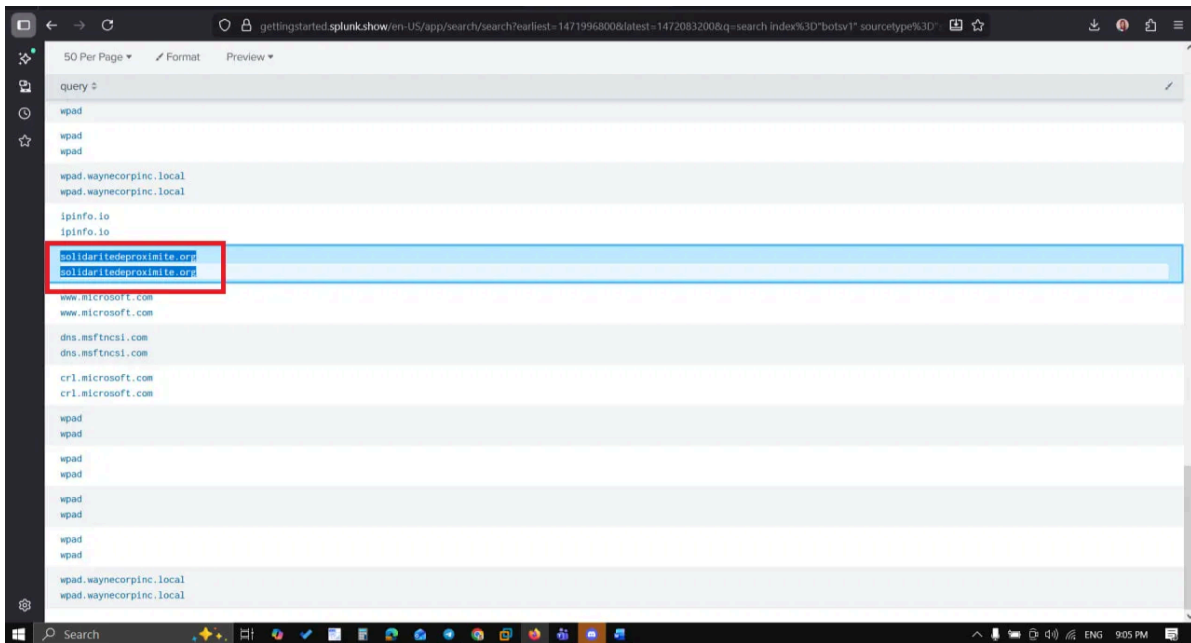
index="botsv1" sourcetype="stream:dns" "*cerber*"



Ans:cerberhhyed5frqa.xmfir0.win

Q4. What was the first suspicious domain visited by we8105desk on 24AUG2016?

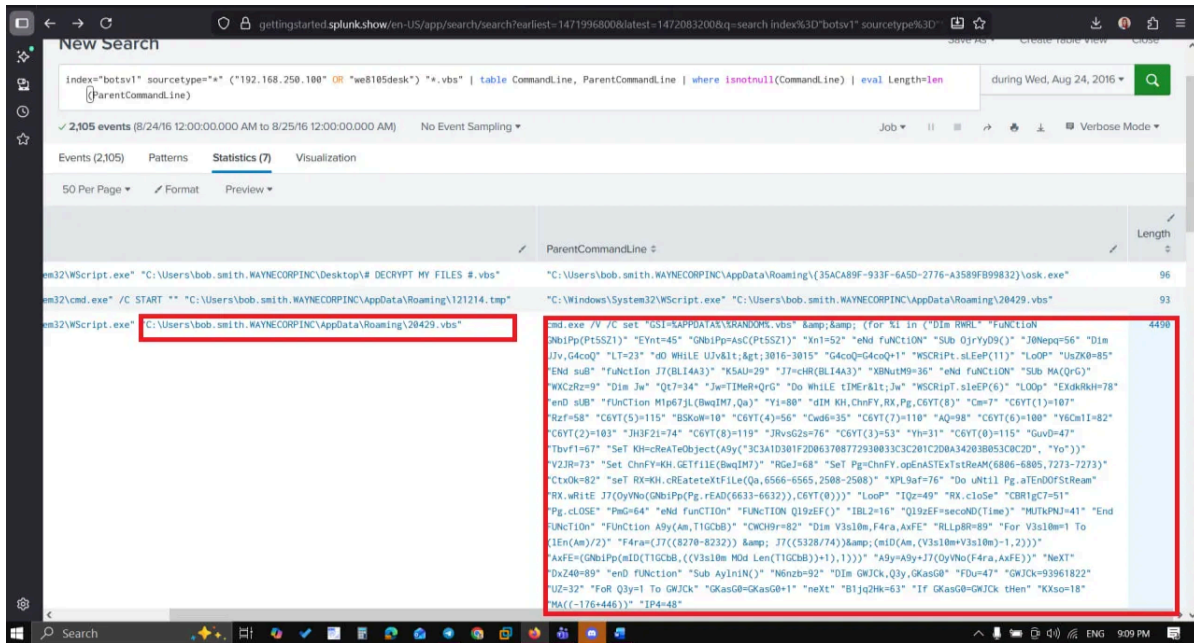
```
index="botsv1" sourcetype="stream:dns" src_ip="192.168.250.100" "query_type{}"="A" | table query
```



Ans: solidaritedeproximitye.org

Q5. During the initial Cerber infection a VB script is run. The entire script from this execution, pre-pended by the name of the launching .exe, can be found in a field in Splunk. What is the length of the value of this field?

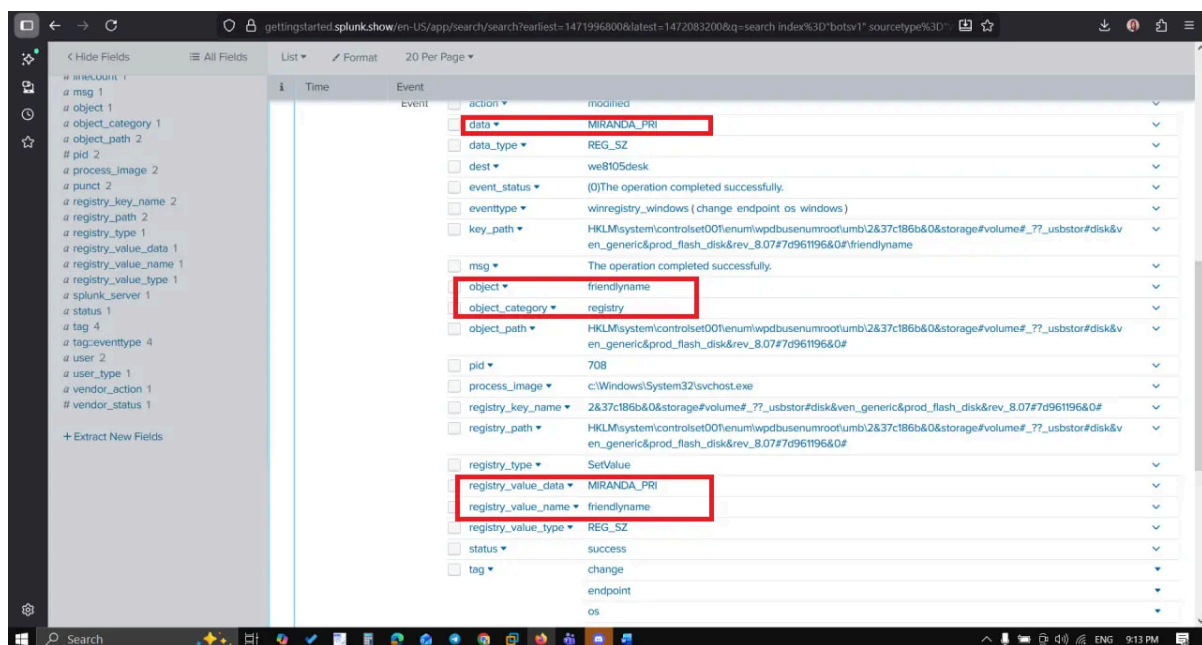
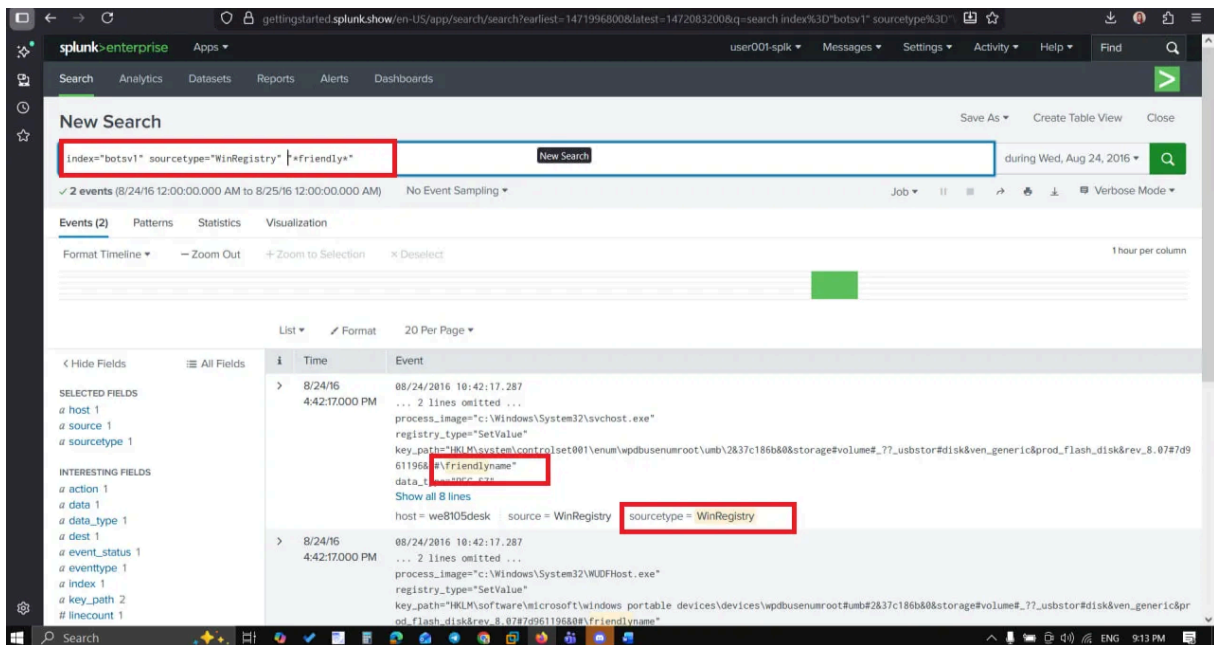
```
index="botsv1" sourcetype="*" ("192.168.250.100" OR "we8105desk") "*.vbs"
| table CommandLine, ParentCommandLine | where isnotnull(CommandLine)
| eval Length=len(ParentCommandLine)
```



Ans:4490

Q6. What is the name of the USB key inserted by Bob Smith?

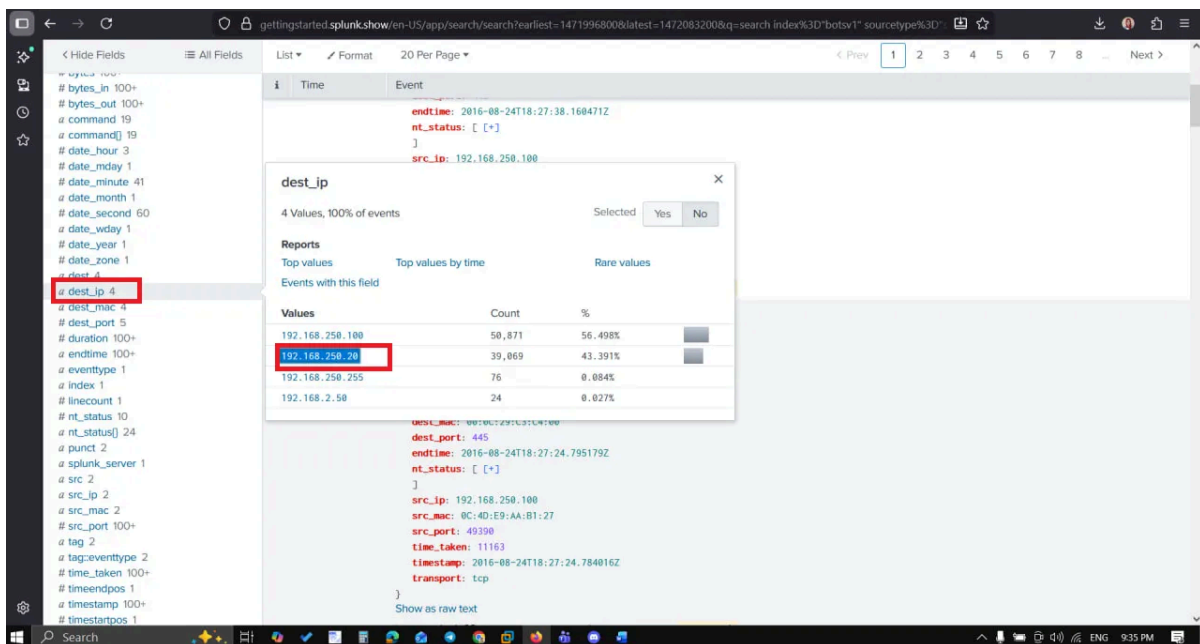
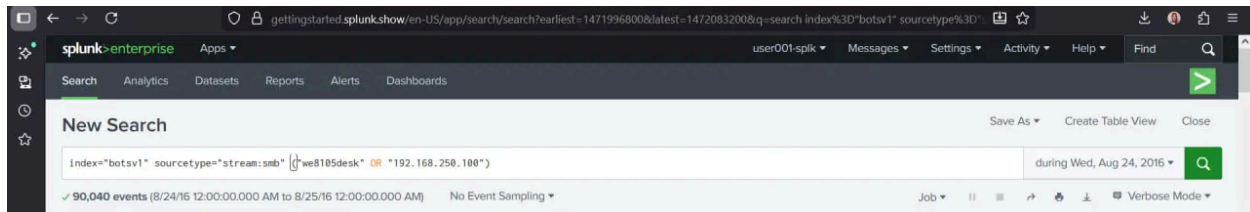
index="botsv1" sourcetype="WinRegistry" "*"friendly*"



Ans:MIRANDA_PRI

Q7. Bob Smith's workstation (we8105desk) was connected to a file server during the ransomware outbreak. What is the IPv4 address of the file server?

index="botsv1" sourcetype="stream:smb" ("we8105desk" OR "192.168.250.100")



Ans:192.168.250.20

Q8. How many distinct PDFs did the ransomware encrypt on the remote file server?

first we need to know the dest hostname

index="botsv1" ("we8105desk" OR "192.168.250.100") dest_port=445

New Search

Index="botsv1" ("we8105desk" OR "192.168.250.100") dest_port=445 dest=we9041srv

25 events (8/24/16 12:00:00.000 AM to 8/25/16 12:00:00.000 AM) No Event Sampling

Events (25) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

1 hour per column

List Format 20 Per Page

< Hide Fields All Fields

SELECTED FIELDS

- a host 1
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- a action 1
- a app 1
- a Computer 1
- a dest 1
- a dest_host 1
- a dest_ip 1
- a direction 1
- a DestinationHostname 1
- a DestinationPort 1
- a DestinationPortName 1

DestinationHostname

1 Value, 100% of events

Selected Yes No

Reports

Top values Top values by time Rare values

Events with this field

Values	Count	%
we9041srv	25	100%

secondly

index=botsv1 host=we9041srv Source_Address="192.168.250.100" "*.pdf"

New Search

Index=botsv1 host=we9041srv Source_Address="192.168.250.100" "*.pdf"

525 events (8/24/16 12:00:00.000 AM to 8/25/16 12:00:00.000 AM) No Event Sampling

Events (525) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

1 hour per column

List Format 20 Per Page

< Hide Fields All Fields

SELECTED FIELDS

- a host 1
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- a Access_Mask 3
- a Accesses 3
- a Account_Name 1
- a action 1
- a app 1
- a AppendData__or_AddSubdirectory__or_CreatePipeInstance__1
- a body 100+
- a ComputerName 1
- a DELETE 1
- a dest 1
- a dest_nt_domain 1
- a dest_nt_host 1

Accesses

3 Values, 100% of events

Selected Yes No

Reports

Top values Top values by time Rare values

Events with this field

Values	Count	%
DELETE	257	48.952%
READ_CONTROL	257	48.952%
ReadAttributes	11	2.095%

Show all 51 lines

host = we9041srv source = WinEventLog:Security sourcetype = WinEventLog:Security

8/24/16 8/24/2016 18:14:55 AM

24 lines omitted ...

Share Name: *\FilesShare

Ans:257

Q9. The VBScript found in question 204 launches 121214.tmp. What is the ParentProcessId of this initial launch?

```
index="botsv1" sourcetype="*" "*121214.tmp*" "/*.vbs"
```

The screenshot shows a Splunk search interface. On the left, a list of fields is displayed, with 'parent_process' selected. On the right, a panel titled 'parent_process' shows the search results. It indicates '1 Value, 100% of events' and provides options to 'Selected' (Yes/No). Below this, there are links for 'Reports' (Top values, Top values by time, Rare values) and 'Events with this field'. A table titled 'Values' shows the following data:

Values	Count	%
C:\Windows\SysWOW64\wscript.exe	1	100%

The screenshot shows a Splunk search interface. On the left, a list of fields is displayed, with 'parent_process_id' selected. On the right, a panel titled 'parent_process_id' shows the search results. It indicates '1 Value, 100% of events' and provides options to 'Selected' (Yes/No). Below this, there are links for 'Reports' (Average over time, Maximum value over time, Minimum value over time, Top values, Top values by time, Rare values) and 'Events with this field'. A table titled 'Values' shows the following data:

Values	Count	%
3968	1	100%

Ans:3968

Q10. The Cerber ransomware encrypts files located in Bob Smith's Windows profile. How many .txt files does it encrypt?

index="botsv1" sourcetype="*" "C:\\Users\\bob.smith*.txt" | dedup TargetFileName

New Search

index="botsv1" sourcetype="*" "C:\\Users\\bob.smith*.txt" host=we8105desk | dedup TargetFileName

410 events (8/24/16 12:00:00.000 AM to 8/25/16 12:00:00.000 AM) No Event Sampling

Source	Count	%
WinEventLog:Microsoft-Windows-Sysmon/Operational	410	100%

New Search

index="botsv1" host=we8105desk sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational" *.txt file_path="C:\\Users\\bob.smith.WAYNECORPINC*" | dedup TargetFileName

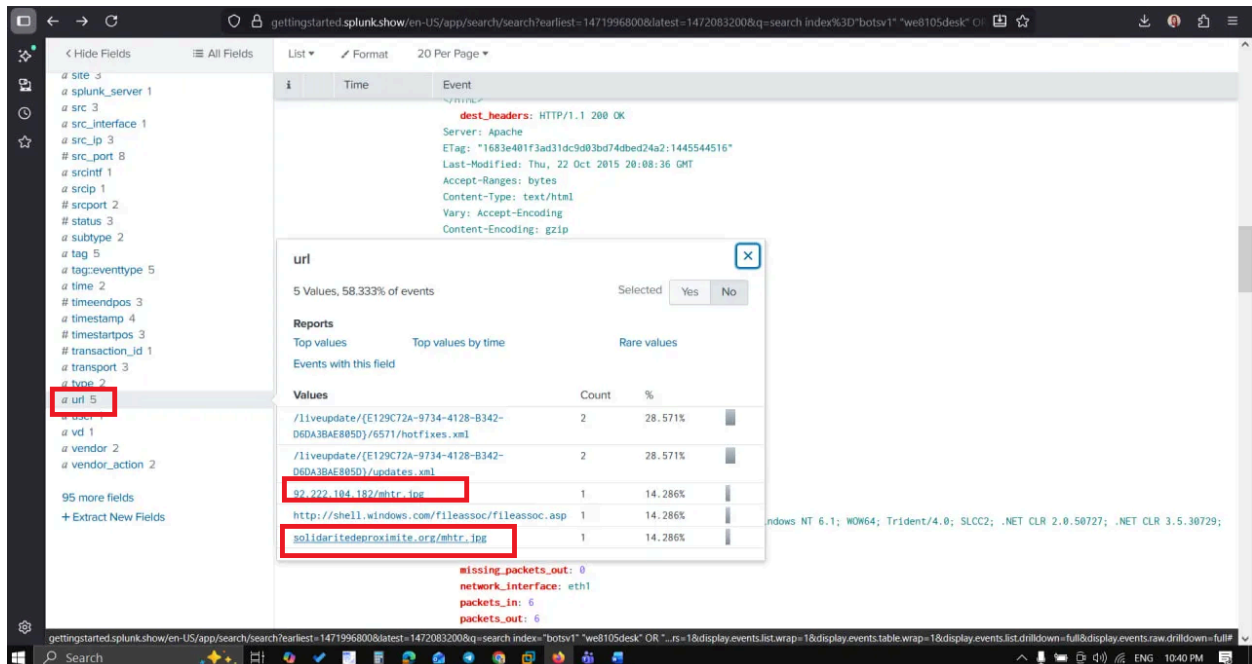
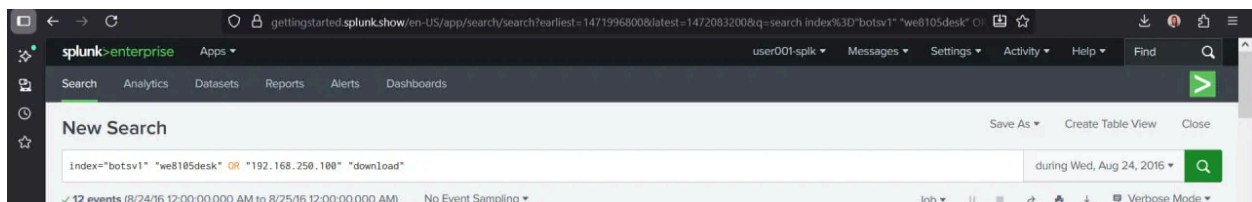
406 events (8/24/16 12:00:00.000 AM to 8/25/16 12:00:00.000 AM) No Event Sampling

Source	Count	%
WinEventLog:Microsoft-Windows-Sysmon/Operational	406	100%

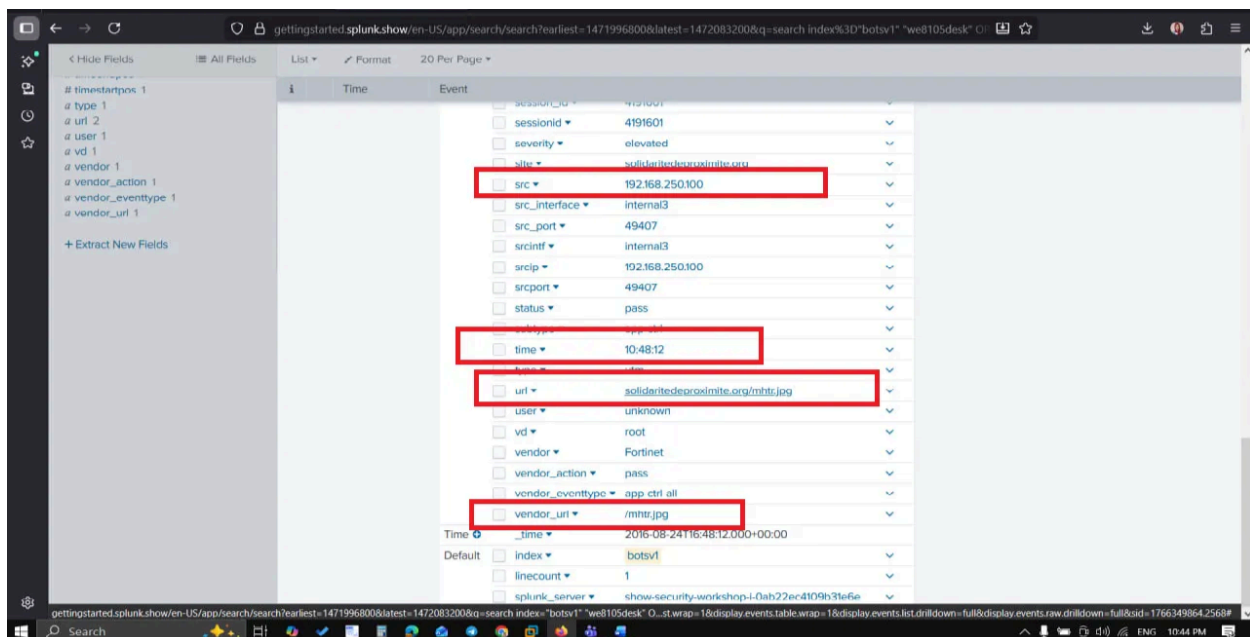
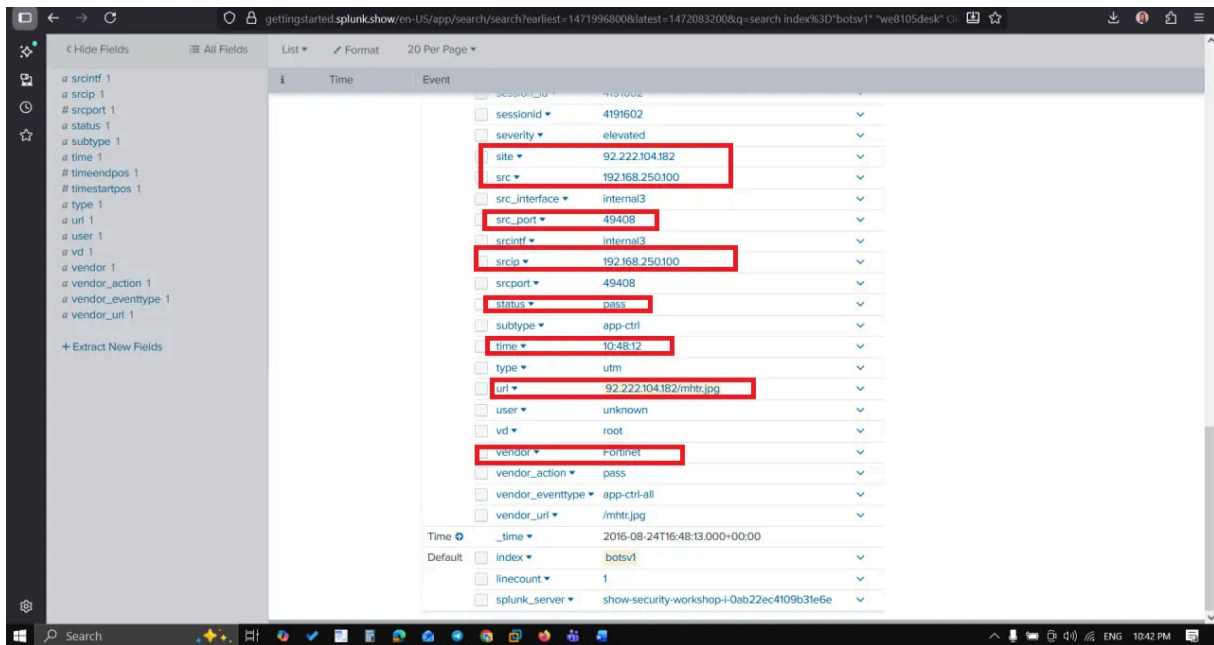
Ans:406

Q11. The malware downloads a file that contains the Cerber ransomware cryptor code. What is the name of that file?

index="botsv1" ("we8105desk" OR "192.168.250.100") "download"



check the two urls details



Ans:mhtr.jpg

Q12. Now that you know the name of the ransomware's encryptor file, what obfuscation technique does it likely use?

Ans:steganography

Explanation: it's a .jpg file (image).

Summary of Indicators of Compromise (IOCs)

- **Victim IP:** 192.168.250.100 \ **hostname:** we8105desk
- **File server IP:** 192.168.250.20 \ **hostname:**we3269srv
- **Infected word document on the USB drive name :**
Miranda_Tate_unveiled.dotm
- **Malicious domain: solidaritedeproximityte.org** and **92.222.104.182**
- **Malicious file downloaded (ransomware cryptor code) :** mhtr.jpg
- **USB device name:** MIRANDA_PRI
- **Suspicious VBScript file:** 121214.tmp

MITRE ATT&CK Mapping

Tactic	Technique
Initial Access	T1204.002 – Malicious File
Execution	T1059 – Command Interpreter
Defense Evasion	T1036 – Masquerading
Command & Control	T1071 – Web Protocols